PART A: VECTOR CONCEPTS

1. Vector Representation

Each student's marks form a vector:

x=[m1,m2,m3,...,mn]x = [m_1, m_2, m_3, ..., m_n]x=[m1,m2,m3,...,mn]

This represents a point in n-dimensional space.

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2. Norms

L1 Norm (Manhattan Distance)

||x||1=∑|xi||x||_1 = \sum |x_i||x||1=∑|xi|

L2 Norm (Euclidean Length)

||x||2=∑xi2||x||_2 = \sqrt{\sum x_i^2}||x||2=∑xi2

Norm measures magnitude of student performance.

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3. Dot Product

x·y=∑xiyix \cdot y = \sum x_i y_ix·y=∑xiyi

Measures similarity between two students.

If large → similar performance
If small → dissimilar

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4. Angle Between Vectors

cosθ=x·y||x||||y||\cos\theta = \frac{x \cdot y}{||x||
||y||}cosθ=||x||||y||x·y

Smaller angle → more similarity.

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5. Projection

projy(x)=x·yy·yyproj_y(x) = \frac{x \cdot y}{y \cdot y} yprojy(x)=y·yx·yy

Shows how much one student's performance aligns with another.

PART B: MATRIX OPERATIONS

A matrix represents the entire dataset:

Xm×nX_{m \times n}Xm×n

Where:

- m = students
- n = subjects

Determinant

Indicates if matrix is invertible.

If determinant ≠ 0 → inverse exists.

Inverse

A−1A^{-1}A−1

Used to solve linear equations.

PART C: COVARIANCE MATRIX

Cov(X)=1n−1XTXCov(X) = \frac{1}{n-1} X^T XCov(X)=n−11XTX

Shows relationships between subjects.

Positive covariance → subjects increase together
Negative covariance → inverse relationship

PART D: EIGENVALUES & EIGENVECTORS

Ax=λxA x = \lambda xAx=λx

Eigenvalues:

- Amount of variance

Eigenvectors:

- Direction of variance

Largest eigenvalue → most important direction.

PART E: LU DECOMPOSITION

A=LUA = LUA=LU

L → Lower triangular
U → Upper triangular

Used for solving systems efficiently.

PART F: SVD

A=UΣVTA = U \Sigma V^TA=UΣVT

- U → row space
- Σ → singular values
- V^T → column space

Used in:

- PCA
- Image compression
- Recommender systems

PART G: PCA

PCA finds directions of maximum variance.

Steps:

1. Standardize data
2. Compute covariance
3. Find eigenvalues
4. Select top components

Reduces dimensionality while keeping maximum information.

PART H: LDA

LDA maximizes class separation.

It:

- Maximizes between-class variance
- Minimizes within-class variance

Used for classification.